Technological Forecasting & Social Change 214 (2025) 124039



Contents lists available at ScienceDirect

Technological Forecasting & Social Change

journal homepage: www.elsevier.com/locate/techfore





Shaping the future through developing and managing breakthrough innovations: A new conceptual framework☆

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ARTICLE INFO

Keywords:
Breakthrough innovations
Emerging technologies
Innovation ecosystems
Digital platforms
Digital businesses
Future society

ABSTRACT

Breakthrough innovations and technologies have gained increasing attention and have the potential to contribute to the development of a better society. Important breakthrough innovations and emerging technologies include fifth generation (5G) telecommunications technologies, artificial intelligence (AI), internet of things, digital platforms and businesses, autonomous vehicles, blockchain, e-payment/e-commerce technologies, and renewable energies. By reviewing recent papers published in *Technological Forecasting and Social Change*, this study proposes a novel conceptual framework for the development and design of such breakthrough technologies. There are three levels in this pyramidal framework: (i) exploring and identifying breakthrough innovations and emerging technologies; (ii) analyzing and managing breakthrough technological innovations and their innovation ecosystems; (iii) synthesizing and strategizing breakthrough innovations such as through digital platforms and businesses. This study makes two important contributions. First, by reviewing a set of coherent studies that shed light on the theoretical and empirical boundaries of research and knowledge in this area, this study pushes these boundaries with the goal of advancing the development and management of breakthrough innovations. Second, this study proposes a novel, integrative conceptual framework which encompasses different levels of activities associated with breakthrough innovations which could help develop and manage breakthrough innovations for the society of tomorrow.

Breakthrough innovations and technologies have been topics of growing interest among academics and managers. They are expected to contribute to the development of a better future society and achievement of sustainable development goals in both developed and emerging economies (Chen et al., 2021; Chen et al., 2020; Huang et al., 2021; Huang and Li, 2019; Miric et al., 2023). We consider such breakthrough technologies to include (inter alia) fifth generation (5G) telecommunications technologies, artificial intelligence (AI), internet of things, digital platforms and businesses, brain-machine interface, autonomous vehicles, blockchain, e-payment/e-commerce technologies, renewable energies, environmental technologies, and space technologies.

This article presents the important means by which global technological capabilities drive the development and realization of future society in both positive and negative ways through global collaboration and competition. It aims to motivate existing players to better develop and utilize breakthrough innovations and strategically manage research and development (R&D) for breakthrough innovations.

In recent years, many incumbent and new companies innovate with emerging and breakthrough technologies to compete in the global arena (Daim et al., 2023; Su and Gibson, 2021). Technological power has shifted due to advancement in breakthrough technologies on a global scale. It has become key to analyzing how the future is shaped by breakthrough innovations and their management, considering the shifting geopolitical, socio-economic and legal landscape and evolving institutions (e.g., Huang et al., 2024a; Huang et al., 2024b; Huang, 2017). Furthermore, many companies with breakthrough innovations,

☆ This article is part of a special issue entitled: Breakthrough Innovations published in Technological Forecasting & Social Change.

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https://doi.org/10.1016/j.techfore.2025.124039
Received 1 February 2025; Accepted 4 February 2025
Available online 18 February 2025
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including emerging technologies, will seek to establish an innovation ecosystem to attain a first-mover advantage and sustain a lasting competitive advantage (Su et al., 2023b; Su et al., 2018a; Su et al., 2018b). In an innovation ecosystem, companies and individuals have incentives to innovate technologically in order to compete or collaborate with other parties, such as governments, suppliers, users, competitors, collaborators, universities, research institutions, etc. (e.g., Chen et al., 2020; Huang et al., 2024b; Jain and Huang, 2022; Jia et al., 2019).

Developing and managing breakthrough innovations requires a coherent strategy which goes beyond technology. Companies can perform better by executing strategies and developing innovations in an efficient and effective manner (Phillips et al., 2019). However, the emergence of breakthrough innovations is realized not only by the strategies and competences of emerging players, but also through other factors. These factors include, beyond the scientific and technological capabilities of these firms and entities, social infrastructure, industrial policy and regulatory reform, business ecosystem-building, social norms and movements, and adaptation under different institutional regimes. They can be theorized from a multi-level perspective and as transition management (Geels, 2002), among other theoretical lenses.

Despite growing scholarly attention to breakthrough innovations and emerging technologies, it is still unclear how existing theories on breakthrough innovations and their management can contribute to the design, development, and realization of a future society consisting of firms/organizations fueled by breakthrough technologies. Prior studies have taken an informetrics approach to design breakthrough technologies (Ogawa and Kajikawa, 2017), identify industrial opportunities (Takano and Kajikawa, 2019), and extract hidden opportunities for those technologies to realize the visions in the future of society (Ittipanuvat et al., 2014). In another study, McDowall (2012) analyzes the transition path and roadmap based on breakthrough technologies and an assessment of the results and impacts of the breakthrough technologies are used to foster collaborations among stakeholders (Kikuchi et al., 2020). However, such design- and action-oriented approaches are scant in the literature on developing and managing breakthrough innovations for society.

To address this gap, we highlight recent papers published in *Technological Forecasting and Social Change*, build upon existing theories and propose a novel framework in the conceptualization and design of a future society enabled by breakthrough technologies. This paper makes two key contributions. First, we review a set of coherent studies to understand the theoretical and empirical boundaries of research and knowledge in this area which allows us to push these boundaries, with the goal of advancing the development and management of breakthrough innovations and technologies. Second, we propose a novel, integrative conceptual framework which encompasses different levels of

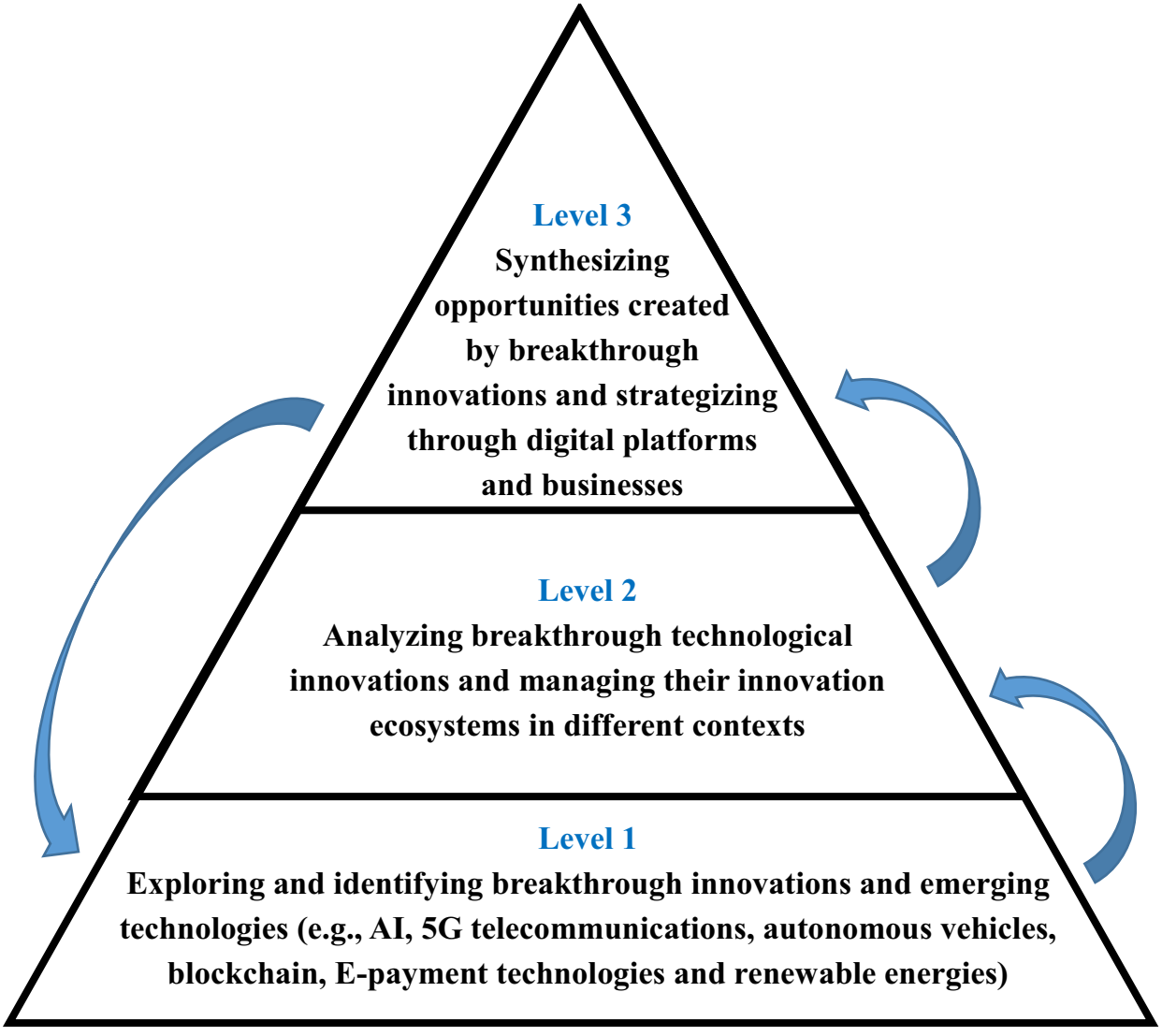


Fig. 1. Pyramidal framework of three levels of key activities and their interactions associated with developing and managing breakthrough innovations.

activities associated with breakthrough innovations. Such framework could help us develop and better manage breakthrough innovations for the society of tomorrow. Specifically, we delineate these three levels of activities in our conceptualization of the pyramidal framework: (i) exploring and identifying breakthrough innovations and emerging technologies; (ii) analyzing breakthrough innovations and managing their innovation ecosystems; (iii) synthesizing opportunities created by breakthrough innovations and strategizing through digital platforms and businesses (See Fig. 1.).

We elaborate on these three levels of important activities associated with breakthrough innovations below:

The first level of activities which most firms undertake focuses on exploring and identifying breakthrough innovations and emerging technologies, such as AI and machine learning (e.g. Miric et al., 2023), 5G vehicle-to-everything (V2X) autonomous vehicles, blockchain, and *E*-Payment technologies, and renewable energies. By reviewing a set of recent studies, we identify the concepts, theories, and methods underlying different breakthrough technologies. We discuss how to explore and develop breakthrough innovations in incumbent companies and startups. Furthermore, we discuss how to manage breakthrough innovations in these companies facing major transformations to adapt to the future society by delineating the processes and mechanisms for managing breakthrough innovations and transitions. We also investigate the key antecedents and consequences during the transition processes, documenting that breakthrough innovations can achieve value creation and capture not only for a single firm or entity, but also across different firms or entities over time. Specifically, in this first level of activities, firms are keen to understand and measure the performance of breakthrough innovations, their impacts under current and future economic conditions, and how to attract investments for such innovations. We propose the important issues in the strategic management of breakthrough innovations, their marketing and supply chains restructuring which could represent a departure from existing businesses.

The second level of activities consists of analyzing breakthrough technological innovations and managing their innovation ecosystems in different contexts (Jacobides et al., 2018; Su et al., 2018a; Su et al., 2018b; Su et al., 2023b). Here, we discuss the mechanisms of ecosystem management and governance among stakeholders for breakthrough innovations and transition management. For examples, how do breakthrough innovations and their innovators and developers interact with their ecosystems and socio-technical regimes and landscape? How do complementarities including niche technologies, social networks, and institutional environments affect breakthrough innovators and their innovations? How do parties, other than end users, in the market, such as international agencies (e.g., United Nations), local governments, and non-governmental organizations (NGOs), participate in and benefit from breakthrough innovations?

The third level of activities, undertaken by the lowest number of firms/organizations, consists of synthesizing opportunities created by breakthrough innovations and strategizing through digital platforms and businesses (Su et al., 2018a; Su et al., 2018b; Su et al., 2023b). Here, we investigate the motivations or inspirations driving breakthrough innovators and key stakeholders in digital platforms and businesses. We discuss how organizations can develop and manage breakthrough innovations supplemented by continuous innovations in an enterprise and/or its digital platform and businesses. We examine how breakthrough innovators can overcome resource constraints and institutional barriers by involving internal and external parties. To do this, it would be important to shed light on the relationships or governance structures related to breakthrough innovations among different stakeholders, e.g., governments, university, research institutes, suppliers, customers, competitors, etc. This advances our understanding of how breakthrough innovators work across the boundaries between markets, industries and countries.

In conceptualizing the three-level of pyramidal framework which would allow us to better understand forces driving development and management of breakthrough innovations, we categorize this special issue's nineteen papers into the three levels of the pyramid. Note there are important interactions and feedback across the levels—from level one to level two, from level two to level three, and from level three to level one.

1. Level 1. Exploring breakthrough innovations and identifying emerging technologies (e.g., AI, 5G telecommunications technologies, 5G-V2X autonomous vehicles, blockchain, *E*-payment technologies and renewable energies)

Firms interested in developing breakthrough technological innovations have to first conduct research in order to identify and understand possible innovations (Datta and Srivastava, 2023), such as AI (Kim et al., 2023), 5G-V2X autonomous vehicles (Su et al., 2023a), automated minibuses (Nemoto et al., 2023), blockchain (Bhatt et al., 2023), e-Payment technologies (Lai et al., 2023) and renewable energies (Buchmann and Wolf, 2023). For level one, we describe these seven studies which explore and identify such breakthrough innovations and emerging technologies.

Datta and Srivastava (2023) conducted a systematic review of the literature and proposed a framework for conceptualizing technological breakthrough innovations. They proposed decision criteria for firms to attain breakthrough innovations by examining the relationship among configurations (i.e., radical versus incremental innovations; architectural versus modular innovations), applications (product versus process innovations; competence-enhancing versus competence-destroying innovations), and performance (breakthrough). This study contributes to the reconceptualization of technological breakthrough innovation.

Nemoto et al. (2023) explored the roles of autonomous vehicles, more specifically, automated minibuses and their integration into the mobility system as a breakthrough technology through the perspective of different stakeholder groups and citizens. They used conceptual mapping and conducted semi-structured interviews with main stakeholder groups ($n = 30$) and a large-scale survey with citizens of four European cities ($n = 1816$). This study contributes to our understanding and identification of the mechanisms for mobility transition with automated minibuses integrated in the mobility systems, supplemented by other niche innovations. Their findings provide support that such systems can be part of the solution that paves the way for a socio-technical transition to a new mobility paradigm.

Su et al. (2023a) assessed the technological trajectory of 5G-V2X autonomous driving inventions using patent analysis. They predicted the growth trend in 5G-V2X technology by using patent analysis and discussed a difference between the number of patents in 5G-V2X technology among the top 10 countries and among the top 10 assignees. They analyzed the 5G-V2X patents in 8 areas—automated vehicle platooning (AVP), cooperative maneuver/advanced driving (CM/AD), extended sensors/cooperative perception (ES/CP), remote driving/teleoperated driving (RD/TD), cooperative safety (CS), autonomous navigation/high definition map (AN/HDM), highway chauffeur (HC), fully automated driving (FAD)—and discussed their distribution in the top 10 countries and top 10 assignees. They also predicted and identified the phase of 5G-V2X technology in technology life cycle.

Bhatt et al. (2023) aimed to identify the technology trajectory and evolution phases of disruptive technology, blockchain, with respect to its predecessor technologies. The data collection was based on keywords and international patent classification (IPC) code to retrieve US patents from Webpat database from the year of 1974 to 2021. They targeted blockchain technology to analyze prior technical knowledge upon which it is derived, as well as how the technologies could diverge or converge to create new applications. This study sheds light on the definitions of five significant technology clusters based on the key-route main path analysis and the evolutionary path of the technology.

Lai et al. (2023) took *E*-payment technology to present the technological development trajectory, and to explore the contribution of

knowledge sources from different regions and various technological field. They used the United States Patent and Trademark Office (USPTO) database as the data source. Analysis of the electronic payment industry is based on the period 1994–2020. The study shows that the E-payment industry technology is dominated by the US. The combination of Qualcomm and Firethorn Mobile holds the leading position in key technologies on the main path of technological development. As well, Visa and Mastercard are major investors in technology R&D to secure their leading position in future technology development.

Kim et al. (2023) argued that explainable AI in recommender systems has surfaced as a solution that can help users understand how and why an AI recommended a specific product or service. They examined the preference for AI interfaces by investigating the user-centered explainability, including scope (global and local) and format (text and visualization). A mixed logit model is used to analyze data collected by a conjoint survey. Results show that local explanation and visualization are preferred, and users dislike lengthy textual interfaces. They find that all four factors are significant for consumers.

Buchmann and Wolf (2024) explored breakthrough inventions in solar PV and wind technologies and the role of scientific discoveries. They identified breakthrough patents based on forward citations. They estimated a firm's breakthrough inventions based on a firm's closeness to science and on other potential driving factors such as the tendency of international cooperation and the average number of involved actors. Furthermore, they compared the solar PV industry (analytic knowledge-base) and wind industry (synthetic knowledge-base) and found that closeness to science is more relevant in the wind industry to create breakthrough patents where technology development mostly relies on engineering knowledge. Furthermore, technological specialization seems more advantageous for breakthrough creation in photovoltaics while breakthrough patents in wind industry correlate with a broader knowledge-base.

2. Level 2. Analyzing breakthrough technological innovations and managing their innovation ecosystems in different contexts

After exploring and identifying breakthrough technological innovations, firms have to analyze and manage these innovations and their ecosystems (Jacobides et al., 2018; Su et al., 2023b; Su et al., 2018a; Su et al., 2018b). For level two, Nylund et al. (2023) analyzed the underlying catalysts for breakthrough innovations ecosystem and proposed crisis as the key driver. Oh et al. (2024) investigated alliance portfolio configuration strategies for managing breakthrough innovations while Wang et al. (2024) unpacked the complementarity in such innovation ecosystems. Kruger and Steyn (2003) examined the university ecosystem in South Africa while Qu et al. (2023) analyzed technological search strategy and breakthrough innovations in China. Yu et al. (2025) studied how breakthrough innovations could emerge and develop in organizations which exhibit multifaceted value logics. We outline these six studies as follows.

Nylund et al. (2023) proposed crisis as catalyst of dormant innovation ecosystems by providing evidence from pandemic surveillance through water. Catalonia of Spain has a successful water ecosystem with one formal water cluster, the Catalan Water Partnership. The Catalan case study shows how previous relationships of trust between agents accelerate new value propositions in the event of a crisis, which functions as a trigger event. They reinforced the concept of an ecosystem as being more of an affiliation than a formal structure. This study contributes to the literature by contrasting idle ecosystems catalyzed by crisis to common value-based ecosystems.

Oh et al. (2024) investigated alliance portfolio configuration strategies as catalysts for innovation using evidence from the international alliances between Japanese and Taiwanese manufacturing companies. They discussed how manufacturing companies strategically configure alliance portfolios to foster innovation over an extended time horizon. They examined resource management and alliance portfolio configuration in Japanese and Taiwanese alliances in automotive and machine tool industries in China. This study contributes to our understanding of the alliance portfolio configuration strategies by shedding new light on several important factors influencing the composition and configuration of long-term alliance portfolios for manufacturing companies.

Wang et al. (2024) unpacked the complementarity in innovation ecosystems and introduced the unique complementarity path and supermodular complementarity path based on prior conceptualization of innovation ecosystems (e.g., Jacobides et al., 2018). They analyzed the impacts of knowledge characteristics (explicit and tacit knowledge), relationship characteristics (trust and tie strength), and complementor characteristics (cultural similarity and absorptive capacity) on breakthrough innovations in these two paths of innovation ecosystems. This study advances our understanding of the factors influencing breakthrough innovations in these two paths of the innovation ecosystems.

Kruger and Steyn (2024) presented a brief overview of the fourth industrial revolution landscape of emerging technologies such as AI, machine learning, big data, blockchain, cloud computing, robotics, internet of things (IoT), additive manufacturing (e.g., 3D printing). Using the diffusion of innovation as a theoretical lens in the case of South Africa, they propose a novel conceptual framework which illustrates how makerspaces and university ecosystem can drive the innovation process and mechanisms in developing and managing breakthrough innovations capabilities. This study also advances our understanding in the context of the emerging economy of South Africa.

Qu et al. (2023) discussed technological search strategy and breakthrough innovation. They analyzed two key measures of breakthrough innovation: (i) technological novelty of firm's performance of breakthrough innovation; and (ii) industrial impact of firm's performance of breakthrough innovation. In this research, both local and distant technology searches can improve these two performance indicators of breakthrough innovations. The study analyzes these two measures of breakthrough innovation and constructs the measures for local and distant search. It also investigates the effectiveness of technological search strategies on managing breakthrough innovations.

Yu et al. (2025) investigated how breakthrough innovations could emerge and develop in organizations which exhibit multifaceted value logics. Under logic multiplicity from the perspective of institutional logic, they discussed the underlying rationale (i.e., logic interplay) of organizational mechanisms that contribute to the emergence and development of breakthrough innovation practices. They conducted in-depth case studies of three breakthrough innovation projects in a prominent Chinese healthcare organization involving multiple logics. Through these studies, a process model for breakthrough innovation development under logic multiplicity is identified. This model contains three sequential mechanisms associated with logic interplay—logic-bridging, logic-harmonizing and logic-tilting. They subsequently contribute to specific breakthrough innovation practices across different stages of the development process. This process model illustrates how organization with multiple logics facilitate the evolution of breakthrough innovations, in which the interplay and coevolution of diverse logics and the agency of organization actors are highlighted.

3. Level 3. Synthesizing opportunities created by breakthrough innovations and strategizing via digital platforms and businesses

In level three, sophisticated firms move beyond analyzing the underlying factors leading to breakthrough innovations. They devote effort and resources to synthesize the opportunities created by these breakthrough innovations and strategize their activities through digital platforms and businesses (Daim et al., 2023; Su et al., 2023b; Su et al., 2018a; Su et al., 2018b). For example, how do firms deploy breakthrough innovation adoption strategies on the business-to-business (B2B) platforms (Caporuscio et al., 2023)? How do the breakthrough technological innovations shape the physical world (Del Vecchio et al.,

2023)? What are the roles of peripheral industrial actors in the development process of digital industrial platform (Su et al., 2023) and the impact of digital economy on firms' breakthrough innovations (Liu et al., 2023)? The firms in this level strategize the roles of technological capabilities when facing digital disruption (Wu et al., 2024) as well as the effects of innovative financial instruments on the development of renewable energy (Song et al., 2023). We delineate these six papers on breakthrough innovations as follows.

Caporuscio et al. (2023) provided an analysis for evaluating the business-to-business (B2B) platforms for breakthrough innovation adoption strategies, enabling better management of future worldwide crises such as the Covid-19 outbreak and the Ukrainian War. They propose a theoretical model based on the most impactful breakthrough innovations on B2B platforms mechanisms. The authors generate four different scenarios for simulation using a dual adaptive management approach. At the end of their simulation, they provided a two-by-two strategy matrix with four main mitigation strategy patterns associated with different levels of crisis perturbation and adaptive approach styles. This study sheds light on the role of digital platforms in configuring an effective crisis mitigation strategy.

Del Vecchio et al. (2023) reviewed the state of the debate on physical technologies or environments. They analyzed the current debate on how digital technologies have been transforming the physical world in the physical environments. They conducted a critical literature review to analyze how breakthrough innovation of physical technologies or environments can reshape the customers' and citizens' experiences. The study highlights the broad spectrum of research interest in physical issues for customers' and citizens' engagement. The adoption strategies of physical technologies for five areas are discussed: i) marketing and consumer behaviour; ii) cultural heritage, tourism and urban development; iii) education and learning; iv) social issues and politics; v) technical and legal issues.

Su et al. (2023) examined the development process of digital industrial platform by peripheral industrial actors. Existing literature has primarily focused on large incumbent firms in developing industrial platforms. For instance, incumbents such as GE, Siemens, Haier, and Sany have established industrial platforms through "platformization" with their respective platforms Predix, MindSphere, COSMOPlat, and RootCloud. This study utilized an exploratory case study of Huaxianbang (HXB), a peripheral actor in China's chemical fabric industry and developed a strategic framework of three digital business model innovations that contribute to successful digital industrial platform developments: i) industrial transaction platformization; ii) industrial factory digitalization; and iii) digital platform enablement.

Liu et al. (2023) studied the impact of digital economy on firms' breakthrough innovations by using patent data of 1728 listed companies in China from 2010 to 2020. They assessed the development of digital economy through these two key aspects: (i) internet development using internet penetration rate, related employees, related output, and mobile phone penetration rate are used; and (ii) digital financial inclusion using the Digital Finance Development Index compiled by Digital Finance Research Center of Peking University and Ant Financial Group (Zou and Deng, 2022). Their findings show that digital economy and digital strategies can significantly enhance breakthrough innovations in Chinese firms in both the short-term and long-term.

Wu et al. (2024) employed an in-depth case study of the Perfect Corporation to investigate the role of technological capabilities in digital disruption. They provided a synthesis of how the market environment, strategies, models, processes, and competencies, and organizational structure influence the company's ability to develop breakthrough innovations successfully. Their findings showed that Perfect Corporation can establish and sustain its competitive advantages by leveraging its parent company's breakthrough technologies and strategically positioning itself in the fashion technology industry to keep abreast of the market trends. This study advances our understanding of how a digital business firm can successfully develop breakthrough innovations under environments with high degree of uncertainty and complexity.

Song et al. (2023) studied the bidirectional causal relationship between innovative financial instruments and the development of renewable energy by using panel data from 30 provinces in China from 2006 to 2018. Song et al. (2023) used an entropy method to construct the explanatory variables such as green finance level indicator—measured by green credit, green investment, green insurance, and governmental support—and dependent variable such as renewable energy—measured by the amount of electricity generated by renewable energy. Three key contributions arise from this study: (i) there is bidirectional causal relationship between green finance and renewable energy development; (ii) there is also significant spatial and temporal correlation between them; (iii) the degree of industrialization, carbon emissions, and population density have a positive influence on renewable energy.

The present overview of the articles offers several managerial implications. First, firms can start by exploring and identifying breakthrough innovations, such as AI, 5G-V2X autonomous vehicles, automated minibuses, blockchain, e-Payment technologies, or renewable energies. Second, firms need to carefully analyze breakthrough innovations to foster the developments of these innovations and manage their ecosystems such as the Catalonia water ecosystem in Spain, manufacturing ecosystems in Taiwan and Japan, and university ecosystem in South Africa. Third, firms can synthesize the opportunities provided by breakthrough innovations, and strategize through digital platforms and businesses, such as the business-to-business (B2B) platforms. The strategic imperatives are to devise effective innovation adoption strategies, enhance breakthrough innovations through the use of digital technologies and platforms, and understand the intricate roles and relationships between firms and their key stakeholders in fostering the development of breakthrough innovations in the era of AI and digital transformation.

CRediT authorship contribution statement

Kenneth G. Huang: Conceptualization, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Yu-Shan Su:** Conceptualization, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jin Chen:** Conceptualization, Writing – original draft, Writing – review & editing. **Yuya Kajikawa:** Conceptualization, Validation, Writing – original draft, Writing – review & editing.

Acknowledgments

Yu-Shan Su thanks the funding support from National Science and Technology Council NSTC111-2410-H-003-041-MY2 and NSTC113-2918-I-003-003-. And Jin Chen thanks the funding support from National Natural Science Foundation of China No. 72232004.

Data availability

No data was used for the research described in the article.

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